

amount of math I had to learn, I could get out of math faster than I could get out of physics. So I ended up getting my PhD in math.

So you were literally almost a double PhD in math and physics.

Exactly.

You never finished your thesis to get a PhD in physics?

No, it wasn't worth my time anymore, although it wouldn't have taken much more time to get it.

Did you like math better than physics?

It was kind of odd. When I was in physics, the way they did things didn't make logical sense to me. They would discuss models of things, but they wouldn't explain the assumptions carefully. I liked the logic of math a lot better. Once I learned the logic of math, I could come back to physics and see quite clearly the assumptions they were making and why they were making them.

As a math professor, how did you get involved in developing blackjack betting systems?

In December 1958 when I was teaching at UCLA for one year before going on to teach at MIT, my wife and I were going to Las Vegas for a low-cost vacation. I knew better than to gamble because the odds are against you. One of the professors in the math department who heard I was going to Las Vegas told me, "There is a new article in the *Journal of the American Statistical Association* that tells you how to play at almost even in blackjack." I thought that if I could play at almost even, I could have some fun without it costing me much. Of course, I didn't think it through all the way because although it might not cost much on average, my experience was not going to be average. It was going to be some random fluctuation around the expected modest negative edge, which could be favorable or unfavorable. I read this article, and it said that I could play with a house edge of only 0.62 percent, which was far better than any other game in Vegas.

I made a blackjack strategy card, and when I got to the tables, I sat down with 10 silver dollars and began to play. People at the table were getting clobbered because the dealer had a really good run of luck. My little